

AIT312 Recommendation System

Previous Question Paper Analysis

PART A Questions (3 Marks Each)

Module 1: Introduction to Basic Concepts and Recent Developments

1. List any two purposes of recommendation system? (2024)
2. How is a collaborative recommendation system categorized? (2024)
3. Indicate the main idea of collaborative recommendation approaches. (2023)
4. Discuss the cases in which content-based recommendations will not perform as well as collaborative filtering. (2023)
5. Explain what is TF-IDF? (2024)

Module 2: Knowledge-based Recommendation

1. Define classical constraint satisfaction problem (CSP). (2023)
2. Differentiate between Dependent default and derived default? (2023, 2024) ⚠️ Repeated

Module 3: Hybrid Recommendation Approaches

1. What are the limitations of hybridization strategies? (2023, 2024) ⚠️ Repeated
2. Explain about feature combination hybrids (2023, 2024) ⚠️ Repeated

Module 4: Evaluating Recommender Systems

1. Differentiate RMSE versus MAE? (2023)
2. Explain evaluation paradigms? (2023, 2024) ⚠️ Repeated
3. How can we differentiate Confidence and Trust? (2024)

Module 5: Attack-Resistant Recommender Systems

1. What is product push attack and nuke attack? (2023, 2024) ⚠️ Repeated
2. Illustrate different methods used to detect attacks on existing recommendation system? (2023)
3. Specify the effect of base recommendation algorithm. (2024)

PART B Questions (14 Marks Each)

Module 1: Introduction to Basic Concepts and Recent Developments

1. Discuss content-based recommendation, in light of explicitly asking the user for his or her interests? (2023)
2. Describe about user-based nearest neighbour recommendation system which deals with new items for which no ratings exist. (2023)
3. Summarize the implicit and explicit rating mechanism in collaborative recommendation approaches. (2023)
4. Describe the two types of outputs generated with pure collaborative approaches that takes matrix of given user-item ratings as the only input? (2024)
5. Explain similarity based retrieval and other text classification methods? (2024)

Module 2: Knowledge-based Recommendation

1. Explain QuickXPlain algorithm that calculates one conflict set at a time for a given set of constraints. (2023, 2024) ⚠️ Repeated
2. Explain about the ranking of items/utility-based recommendation. (2023)
3. Describe Critiquing algorithms. (2023, 2024) ⚠️ Repeated
4. How can we deal un-satisfiable requirements and empty results sets? (2024)

Module 3: Hybrid Recommendation Approaches

1. Explain recommendation paradigm? (2023)
2. Describe feature augmentation hybrid. (2023, 2024) ⚠️ Repeated
3. Explain about different parallelized hybridization strategies. (2023, 2024) ⚠️ Repeated
4. Differentiate between cascade hybrids and meta-level hybrids? (2023, 2024) ⚠️ Repeated
5. Explain about feature combination hybrids. (2024)

Module 4: Evaluating Recommender Systems

1. Discuss about the general goals of evaluation design? (2023, 2024) ⚠️ Repeated
2. Discuss about the design issues in offline recommender evaluation. Illustrate with a case study. (2023, 2024) ⚠️ Repeated
3. Explain about accuracy metrics in offline evaluation. (2023, 2024) ⚠️ Repeated
4. Explain about offline and online evaluations in recommender systems. (2024)

Module 5: Attack-Resistant Recommender Systems

1. Discuss about different attacks on recommender system? (2023, 2024) ⚠️ Repeated
2. How do you quantify attack impact on recommender system? (2023, 2024) ⚠️ Repeated
3. Discuss about different methods available to detect attacks on recommender system (2024)
4. Explain how to design robust recommendation algorithms. (2024)

Summary of Repeated Questions

Most Important Questions (Repeated in Both Years)

1. **Differentiate between Dependent default and derived default?** (Part A)
2. **What are the limitations of hybridization strategies?** (Part A)
3. **Explain about feature combination hybrids** (Part A)
4. **Explain evaluation paradigms?** (Part A)
5. **What is product push attack and nuke attack?** (Part A)
6. **Explain QuickXPlain algorithm that calculates one conflict set at a time for a given set of constraints.** (Part B)
7. **Describe Critiquing algorithms.** (Part B)
8. **Describe feature augmentation hybrid.** (Part B)
9. **Explain about different parallelized hybridization strategies.** (Part B)
10. **Differentiate between cascade hybrids and meta-level hybrids?** (Part B)
11. **Discuss about the general goals of evaluation design?** (Part B)
12. **Discuss about the design issues in offline recommender evaluation. Illustrate with a case study.** (Part B)
13. **Explain about accuracy metrics in offline evaluation.** (Part B)
14. **Discuss about different attacks on recommender system?** (Part B)
15. **How do you quantify attack impact on recommender system?** (Part B)

Important Topics by Module

Module 1: Introduction to Basic Concepts and Recent Developments

- Collaborative recommendation approaches
- User-based vs Item-based nearest neighbor recommendation
- Content-based recommendation, TF-IDF
- Implicit and explicit rating mechanisms
- Text classification methods

Module 2: Knowledge-based Recommendation

- Constraint satisfaction problems (CSP)
- Dependent vs derived defaults
- QuickXPlain algorithm
- Dealing with unsatisfiable requirements

- Utility-based recommendation
- Critiquing algorithms

Module 3: Hybrid Recommendation Approaches

- Feature combination hybrids
- Feature augmentation hybrids
- Parallelized hybridization strategies
- Cascade hybrids vs meta-level hybrids
- Limitations of hybridization strategies

Module 4: Evaluating Recommender Systems

- Evaluation paradigms
- Offline vs Online evaluation
- RMSE vs MAE metrics
- Confidence and Trust
- Design issues in offline evaluation
- Accuracy metrics

Module 5: Attack-Resistant Recommender Systems

- Product push and nuke attacks
- Methods to detect attacks
- Quantifying attack impact
- Effect of base recommendation algorithm
- Robust recommendation algorithms

Exam Weightage Analysis

Module	No. of Questions	Percentage
Module 1	7	23%
Module 2	6	19%
Module 3	7	23%
Module 4	7	23%
Module 5	7	23%

The exam appears to have an even distribution across all five modules, with a slightly higher focus on Modules 1, 3, 4, and 5.

Conclusion

Based on the analysis of previous question papers, the most important areas to focus on are the repeated questions, particularly those in Part B which carry more marks. The hybrid recommendation approaches, evaluation methods, and attack-resistant systems seem to have more repeated questions, suggesting a higher emphasis on these topics.